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SINGLE-INPUT DUAL-OUTPUT ANALOG-TO-DIGITAL CONVERTER

Abstract

A single-input dual-output (SIDO) analog-to-digital converter (ADC) is disclosed. Herein, the SIDO ADC is configured to receive an analog input voltage and concurrently output a digital average and a digital peak of the received analog input voltage. Moreover, the SIDO ADC can be configured with a configurable dynamic range to output the digital average and the digital peak with sufficient granularity when the analog input voltage is associated with a larger peak-to-average ratio (PAR). As such, the SIDO ADC can effectively overcome the limitations in an existing single-input single-output (SISO) ADC.

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Background/Summary

RELATED APPLICATIONS [0001] This application claims the benefit of U.S. provisional patent application Ser. No. 63/557,675, filed on Feb. 26, 2024, the disclosure of which is hereby incorporated herein by reference in its entirety.

FIELD OF THE DISCLOSURE

[0002] The technology of the disclosure relates generally to a single-input dual-output (SIDO) analog-to-digital converter (ADC) that can convert concurrently an analog input signal into a digital average signal and a digital peak signal.

BACKGROUND

[0003] Mobile communication devices have become increasingly common in current society. The prevalence of these mobile communication devices is driven in part by the many functions that are now enabled on such devices. Increased processing capabilities in such devices means that mobile communication devices have evolved from being pure communication tools into sophisticated mobile multimedia centers that enable enhanced user experiences.

[0004] The redefined user experience requires higher data rates offered by wireless communication technologies, such as long-term evolution (LTE) and fifth-generation new radio (5G-NR). To achieve the higher data rates in mobile communication devices, sophisticated power amplifiers (PAs) may be employed to increase output power of radio frequency (RF) signals (e.g., maintaining sufficient energy per bit) communicated by mobile communication devices. However, the increased output power of RF signals can lead to increased power consumption and thermal dissipation in mobile communication devices, thus compromising overall performance and user experiences.

[0005] Notably, a mobile communication device often includes many analog-to-digital converters (ADCs) to convert analog signals into digital signals. FIG. 1 is a schematic diagram of an exemplary single-input single-output (SISO) ADC **10** that can convert an analog voltage $V_{sub.A}$ into a digital average voltage $V_{sub.AVG}$. For an in-depth description as to how the existing SISO ADC **10** can convert the analog voltage $V_{sub.A}$ to the digital average voltage $V_{sub.AVG}$, please refer to U.S. Pat. No. 11,268,990 B2, entitled “CURRENT MEASUREMENT CIRCUIT FOR OPTIMIZATION OF POWER CONSUMPTION IN ELECTRONIC DEVICES.”

[0006] The existing SISO ADC **10** may have certain limitations. In one aspect, the existing SISO ADC **10** may have a limited dynamic range that may cause the digital average voltage $V_{sub.AVG}$ to have a lower granularity when the analog voltage $V_{sub.A}$ is associated with a large peak-to-average ratio (PAR). In another aspect, although the existing SISO ADC **10** can produce the digital average voltage $V_{sub.AVG}$, the existing SISO ADC **10** is unable to capture a digital peak of the analog voltage $V_{sub.A}$. As such, it is desirable to optimize the existing SISO ADC **10** to overcome the above-mentioned limitations.

SUMMARY

[0007] Embodiments of the disclosure relate to a single-input dual-output (SIDO) analog-to-digital converter (ADC). Herein, the SIDO ADC is configured to receive an analog input voltage and concurrently output a digital average and a digital peak of the received analog input voltage. Moreover, the SIDO ADC can be configured with a configurable dynamic range to output the digital average and the digital peak with sufficient granularity when the analog input voltage is associated with a larger peak-to-average ratio (PAR). As such, the SIDO ADC can effectively overcome the limitations in an existing single-input single-output (SISO) ADC.

[0008] In one aspect, a SIDO ADC is provided. The SIDO ADC includes a scaling circuit. The

scaling circuit is configured to receive an analog input voltage and scale the analog input voltage down to a sense voltage. The SIDO ADC also includes a digital average voltage circuit. The digital average voltage circuit is configured to store a baseline voltage and multiple threshold voltages each higher than the baseline voltage to thereby establish multiple threshold regions. The digital average voltage circuit is also configured to count each occurrence of the sense voltage being higher than or equal to a highest one of the plurality of threshold voltages that falls below the sense voltage during a predefined measurement period in a corresponding one of multiple first counters. The digital average voltage circuit is also configured to reduce the sense voltage by a respective one of multiple offset values corresponding to the highest one of the plurality of threshold voltages that falls below the sense voltage. The digital average voltage circuit is also configured to output a digital average value indicating an average of the analog input voltage during the predefined measurement period based on a respective value in each of the multiple first counters. The SIDO ADC also includes a digital peak voltage circuit. The digital peak voltage circuit is configured to count a respective duration of the sense voltage staying within each of the multiple threshold regions during the predefined measurement period in multiple second counters. The digital peak voltage circuit is also configured to output a digital peak value indicating a peak of the analog input voltage during the predefined measurement period based on a respective value in each of the multiple first counters and each of the multiple second counters.

[0009] In another aspect, a method for converting an analog input voltage into a digital average value and a digital peak value is provided. The method includes receiving the analog input voltage and scaling the analog input voltage down to a sense voltage. The method also includes storing a baseline voltage and multiple threshold voltages each higher than the baseline voltage to thereby establish multiple threshold regions. The method also includes counting each occurrence of the sense voltage being higher than or equal to a highest one of the plurality of threshold voltages that falls below the sense voltage during a predefined measurement period in a corresponding one of multiple first counters. The method also includes reducing the sense voltage by a respective one of multiple offset values corresponding to the highest one of the plurality of threshold voltages that falls below the sense voltage. The method also includes outputting the digital average value indicating an average of the analog input voltage during the predefined measurement period based on a respective value in each of the multiple first counters. The method also includes counting a respective duration of the sense voltage staying within each of the multiple threshold regions during the predefined measurement period in multiple second counters. The method also includes outputting the digital peak value indicating a peak of the analog input voltage during the predefined measurement period based on a respective value in each of the multiple first counters and each of the multiple second counters.

[0010] In another aspect, a wireless device is provided. The wireless device includes at least one SIDO ADC. The at least one SIDO ADC includes a scaling circuit. The scaling circuit is configured to receive an analog input voltage and scale the analog input voltage down to a sense voltage. The at least one SIDO ADC also includes a digital average voltage circuit. The digital average voltage circuit is configured to store a baseline voltage and multiple threshold voltages each higher than the baseline voltage to thereby establish multiple threshold regions. The digital average voltage circuit is also configured to count each occurrence of the sense voltage being higher than or equal to a highest one of the plurality of threshold voltages that falls below the sense voltage during a predefined measurement period in a corresponding one of multiple first counters. The digital average voltage circuit is also configured to reduce the sense voltage by a respective one of multiple offset values corresponding to the highest one of the plurality of threshold voltages that falls below the sense voltage. The digital average voltage circuit is also configured to output a digital average value indicating an average of the analog input voltage during the predefined measurement period based on a respective value in each of the multiple first counters. The at least one SIDO ADC also includes a digital peak voltage circuit. The digital peak voltage circuit is

configured to count a respective duration of the sense voltage staying within each of the multiple threshold regions during the predefined measurement period in multiple second counters. The digital peak voltage circuit is also configured to output a digital peak value indicating a peak of the analog input voltage during the predefined measurement period based on a respective value in each of the multiple first counters and each of the multiple second counters.

[0011] Those skilled in the art will appreciate the scope of the present disclosure and realize additional aspects thereof after reading the following detailed description of the preferred embodiments in association with the accompanying drawing figures.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

[0012] The accompanying drawing figures incorporated in and forming a part of this specification illustrate several aspects of the disclosure, and together with the description serve to explain the principles of the disclosure.

[0013] FIG. 1 is a schematic diagram of an existing single-input single-output (SISO) analog-to-digital converter (ADC) having some specific limitations;

[0014] FIG. 2 is a schematic diagram of an exemplary single-input dual-output (SIDO) ADC configured to output a digital average and a digital peak of an analog input voltage;

[0015] FIG. 3 is a schematic diagram providing an exemplary illustration of the SIDO ADC of FIG. 2 configured according to an embodiment of the present disclosure to output the digital average and the digital peak of the analog input voltage;

[0016] FIG. 4 is a graphic diagram providing an exemplary graphical illustration of multiple threshold voltages and multiple threshold regions whereby the SIDO ADC of FIG. 3 can output the digital average and the digital peak of the analog input voltage;

[0017] FIG. 5 is a graphic diagram illustrating a working example of the SISO ADC of FIG. 3;

[0018] FIG. 6 is a schematic diagram of an exemplary user element wherein the SIDO ADC of FIG. 2 can be provided; and

[0019] FIG. 7 is a flowchart of an exemplary process whereby the SIDO ADC of FIG. 2 can output the digital average and the digital peak of the analog input voltage.

DETAILED DESCRIPTION

[0020] The embodiments set forth below represent the necessary information to enable those skilled in the art to practice the embodiments and illustrate the best mode of practicing the embodiments. Upon reading the following description in light of the accompanying drawing figures, those skilled in the art will understand the concepts of the disclosure and will recognize applications of these concepts not particularly addressed herein. It should be understood that these concepts and applications fall within the scope of the disclosure and the accompanying claims.

[0021] It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the present disclosure. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0022] It will be understood that when an element such as a layer, region, or substrate is referred to as being “on” or extending “onto” another element, it can be directly on or extend directly onto the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” or extending “directly onto” another element, there are no intervening elements present. Likewise, it will be understood that when an element such as a layer, region, or substrate is referred to as being “over” or extending “over” another element, it can be directly over

or extend directly over the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly over” or extending “directly over” another element, there are no intervening elements present. It will also be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present.

[0023] Relative terms such as “below” or “above” or “upper” or “lower” or “horizontal” or “vertical” may be used herein to describe a relationship of one element, layer, or region to another element, layer, or region as illustrated in the Figures. It will be understood that these terms and those discussed above are intended to encompass different orientations of the device in addition to the orientation depicted in the Figures.

[0024] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises,” “comprising,” “includes,” and/or “including” when used herein specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0025] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms used herein should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0026] Embodiments of the disclosure relate to a single-input dual-output (SIDO) analog-to-digital converter (ADC). Herein, the SIDO ADC is configured to receive an analog input voltage and concurrently output a digital average and a digital peak of the received analog input voltage. Moreover, the SIDO ADC can be configured with a configurable dynamic range to output the digital average and the digital peak with sufficient granularity when the analog input voltage is associated with a larger peak-to-average ratio (PAR). As such, the SIDO ADC can effectively overcome the limitations in an existing single-input single-output (SISO) ADC.

[0027] FIG. 2 is a schematic diagram of an exemplary SIDO ADC **12** configured to output a digital average $V_{\text{sub.AVG}}$ and a digital peak $V_{\text{sub.PEAK}}$ of an analog input voltage $V_{\text{sub.A}}$. As described in detail below, the SIDO ADC **12** can be configured with a configurable dynamic range to output the digital average $V_{\text{sub.AVG}}$ and the digital peak $V_{\text{sub.PEAK}}$ with sufficient granularity, even when the analog input voltage $V_{\text{sub.A}}$ is associated with a larger PAR. As such, the SIDO ADC **12** can effectively overcome the limitations in the SISO ADC **10** of FIG. 1.

[0028] FIG. 3 is a schematic diagram of the SIDO ADC **12** of FIG. 2 configured according to an embodiment of the present disclosure. Common elements between FIGS. 2 and 3 are shown therein with common element numbers and will not be re-described herein.

[0029] Herein, the SIDO ADC **12** includes a scaling circuit **14**, a digital average voltage circuit **16**, and a digital peak voltage circuit **18**. The scaling circuit **14** is configured to receive the analog input voltage $V_{\text{sub.A}}$, which is associated with an electrical current $I_{\text{sub.actual}}$, and scale down the analog input voltage $V_{\text{sub.A}}$ to a sense voltage $V_{\text{sub.sense}}$ that is lower than the analog input voltage $V_{\text{sub.A}}$. One benefit of scaling down the analog input voltage $V_{\text{sub.A}}$ is that the digital average voltage circuit **16** and the digital peak voltage circuit **18** will each operate with a lower voltage, thus allowing these circuits to be built with smaller electrical components (e.g., transistors, capacitors, etc.) to help reduce footprint and cost of the SIDO ADC **12**.

[0030] In an embodiment, the scaling circuit **14** includes a voltage-to-current converter **20**, a

capacitor **22**, and a reference current generator **24**. The voltage-to-current converter **20** is configured to convert the analog input voltage $V_{sub.A}$ to a sense current $I_{sub.sense}$ that is proportionally related to the electrical current $I_{sub.actual}$ as shown in equation (Eq. 1) below.

[00001] $I_{sense} = I_{actual} / C_{ratio}$ (Eq. 1)

[0031] In the equation (Eq. 1) above, $C_{sub.ratio}$ ($C_{sub.ratio} > 1$) represents a scaling factor (a.k.a. a ratio between the electrical current $I_{sub.actual}$ and the sense current $I_{sub.sense}$). The capacitor **22**, which has a capacitance C_O , is continuously charged by the sense current $I_{sub.sense}$ to generate the sense voltage $V_{sub.sense}$. Understandably, since the sense current $I_{sub.sense}$ is related to the electrical current $I_{sub.actual}$ by the scaling factor $C_{sub.ratio}$, the sense voltage $V_{sub.sense}$ is likewise related to the analog input voltage $V_{sub.A}$ by the scaling factor $C_{sub.ratio}$. Thus, it is possible to convert the analog input voltage $V_{sub.A}$ into the digital average value $V_{sub.AVG}$ and the digital peak value $V_{sub.PEAK}$ based on the sense voltage $V_{sub.sense}$.

[0032] The reference current generator **24** may be coupled to or decoupled from the capacitor **22** by closing or opening a switch SW. When the reference current generator **24** is coupled to the capacitor **22**, the reference current generator **24** will generate a reference current $I_{sub.ref}$ that flows in an opposite direction relative to the sense current $I_{sub.sense}$. In this regard, the reference current $I_{sub.ref}$ may be combined with the sense current $I_{sub.sense}$ to generate a combined current $I_{sub.sum}$ ($I_{sub.sum} = I_{sub.sense} - I_{sub.ref}$). Since the reference current $I_{sub.ref}$ flows in the opposite direction relative to the sense current $I_{sub.sense}$, the reference current $I_{sub.ref}$ will offset (a.k.a. reduce) the sense current $I_{sub.sense}$ that flows through the capacitor **22**. As a result, the combined current $I_{sub.sum}$ may become smaller to thereby reduce the sense voltage $V_{sub.sense}$. Understandably, by generating the reference current $I_{sub.ref}$ at different levels, it is possible to reduce the sense voltage $V_{sub.sense}$ by different offset values $OFF_{sub.1}$ - $OFF_{sub.N}$. As further described below, the ability to reduce the sense voltage $V_{sub.sense}$ by different offset values $OFF_{sub.1}$ - $OFF_{sub.N}$ allows the SIDO ADC **12** to have a configurable dynamic range and sufficient granularity when the analog input voltage $V_{sub.A}$ is associated with a larger PAR.

[0033] In an embodiment, the digital average voltage circuit **16** includes a range detection logic **26**, multiple voltage comparators **28(1)-28(N)**, multiple first latches **30(1)-30(N)**, multiple first counters **32(1)-32(N)** (e.g., flip-flop counters), and an average calculation circuit **34**. Herein, there exists a one-to-one correspondence (a.k.a. association) between the voltage comparators **28(1)-28(N)**, the first latches **30(1)-30(N)**, and the first counters **32(1)-32(N)**. As an example, the voltage comparator **28(1)** corresponds to the first latch **30(1)** and the first counter **32(1)**, the voltage comparator **28(2)** corresponds to the first latch **30(2)** and the first counter **32(2)**, and so on. The range detection logic **26** and the first latches **30(1)-30(N)** are each configured to operate based on a clock signal Clk, which corresponds to a number of clock cycles **36**. Each of the clock cycles **36** has a respective clock duration $T_{sub.clk}$.

[0034] Each of the voltage comparators **28(1)-28(N)** is configured to compare the sense voltage $V_{sub.sense}$ with a respective one of multiple threshold voltages $V_{sub.1}$ - $V_{sub.N}$ to determine whether the sense voltage $V_{sub.sense}$ is higher than or equal to the respective one of the threshold voltages $V_{sub.1}$ - $V_{sub.N}$. In an embodiment, the threshold voltages $V_{sub.1}$ - $V_{sub.N}$ are different from one another and arranged in an ascending order ($V_{sub.1} < V_{sub.2} < \dots < V_{sub.N}$). FIG. 4 provides an exemplary illustration of the threshold voltages $V_{sub.1}$ - $V_{sub.N}$ and their correspondence to the voltage comparators **28(1)-28(N)**.

[0035] Herein, $V_{sub.0}$ represents a baseline voltage (a.k.a. initial voltage) of the sense voltage $V_{sub.sense}$ and is lower than each of the threshold voltages $V_{sub.1}$ - $V_{sub.N}$ ($V_{sub.0} < V_{sub.1}$ - $V_{sub.N}$). The baseline voltage $V_{sub.0}$ and the threshold voltage $V_{sub.1}$ as well as each adjacent pair of the threshold voltages $V_{sub.1}$ - $V_{sub.N}$ can define a respective one of multiple threshold regions $R_{sub.1}$ - $R_{sub.N}$. For example, the baseline voltage $V_{sub.0}$ and the threshold voltage $V_{sub.1}$ collectively define the threshold region $R_{sub.1}$, the adjacent pair of the threshold voltages

V.sub.1 and V.sub.2 collectively define the threshold region R2, and so on. Although the SIDO ADC **12** is indifferent to how the threshold regions R.sub.1-R.sub.N are defined, it may be preferable to have the threshold regions R.sub.1-R.sub.N equally spaced.

[0036] With reference back to FIG. 3, the threshold voltages V.sub.1-V.sub.N may be determined in accordance with the PAR of the analog input voltage V.sub.A and/or a desired granularity in the digital average value V.sub.AVG and the digital peak value V.sub.PEAK. Thus, by determining the threshold voltages V.sub.1-V.sub.N properly, the SIDO ADC **12** can be configured to support a configurable dynamic range with desirable granularity. In one embodiment, the threshold voltages V.sub.1-V.sub.N may be configured statically (e.g., by preloading into onboard registers). In another embodiment, the threshold voltages V.sub.1-V.sub.N may be configured dynamically (e.g., via a radio frequency frontend (RFFE) interface).

[0037] Herein, each of the voltage comparators **28(1)-28(N)** is further configured to output a respective threshold crossing indication IND.sub.i ($1 \leq i \leq N$) in response to determining that the sense voltage V.sub.sense is higher than or equal to the respective one of the threshold voltages V.sub.1-V.sub.N. Notably, not everyone of the voltage comparators **28(1)-28(N)** will output the respective threshold crossing indication IND.sub.i at a given time. For example, if the sense voltage V.sub.sense is higher than or equal to the threshold voltage V.sub.2, then only the voltage comparators **28(1)** and **28(2)** will output the respective threshold crossing indication IND.sub.1 and IND.sub.2. Understandably, the voltage comparators **28(1)-28(N)** will only output the respective threshold crossing indication IND.sub.i simultaneously when the sense voltage V.sub.sense is higher than or equal to the threshold voltage V.sub.N. In contrast, none of the voltage comparators **28(1)-28(N)** will output the respective threshold crossing indication IND.sub.i if the sense voltage V.sub.sense is below the threshold voltage V.sub.1.

[0038] The range detection logic **26** is configured to determine which of the respective threshold crossing indications IND.sub.i-IND.sub.N is provided by a respective one of the voltage comparators **28(1)-28(N)** corresponding to a highest one of the threshold voltages V.sub.1-V.sub.N that has been crossed by the sense voltage V.sub.sense. For example, if both the voltage comparators **28(1)** and **28(2)** output the respective threshold crossing indication IND.sub.1 and IND.sub.2, then the range detection logic **26** will determine that the respective threshold crossing indication IND.sub.2 is associated with the highest threshold voltage V.sub.2 among the threshold voltages V.sub.1 and V.sub.2 that have been crossed by the sense voltage V.sub.sense. Accordingly, the range detection logic **26** can provide a latch signal **40** to a corresponding one of the first latches **30(1)-30(N)** to thereby cause a corresponding one of the first counters **32(1)-32(N)** to increase by one (1). In the earlier example where the range detection logic **26** determines that the threshold voltage V.sub.2 is the highest one among the threshold voltages V.sub.1 and V.sub.2, the range detection logic **26** will provide the latch signal **40** to the first latch **30(2)** to thereby cause the first counter **32(2)** to increase by 1.

[0039] Concurrent or subsequent to providing the latch signal **40**, the range detection logic **26** further provides a control signal **42** to close the switch SW to thereby couple the reference current generator **24** to the capacitor **22**. As described earlier, the reference current generator **24** will generate the reference current I.sub.ref that flows in the opposite direction relative to the sense current I.sub.sense to reduce the sense current I.sub.sense and thereby the sense voltage V.sub.sense to below the highest one of the threshold voltages V.sub.1-V.sub.N. Referring to the above example once again, the range detection logic **26** will close the switch SW such that the sense voltage V.sub.sense can be reduced to below the threshold voltage V.sub.2. Depending on the amount of the reference current I.sub.ref being generated, the sense voltage V.sub.sense may even be reduced to below the threshold voltage V.sub.1. In an embodiment, the amount of the reference current I.sub.ref that the reference current generator **24** will generate can be defined by equation (Eq. 2) below.

$$[00002] I_{\text{ref}}(i) = i * (N_{\text{max}} * I_{\text{unit}} / C_{\text{ratio}}) (1 \leq i \leq N) \quad (\text{Eq. 2})$$

[0040] Herein, i represents an index of the threshold voltages $V_{\text{sub.1}}-V_{\text{sub.N}}$ ($1 \leq i \leq N$). $I_{\text{sub.ref}}(i)$ represents the reference current $I_{\text{sub.ref}}$ to be generated when the threshold voltage $V_{\text{sub.i}}$ ($1 \leq i \leq N$) among the threshold voltages $V_{\text{sub.1}}-V_{\text{sub.N}}$ is determined to be the highest threshold voltage being crossed by the sense voltage $V_{\text{sub.sense}}$. Once again using the example described earlier, when the threshold voltage $V_{\text{sub.2}}$ is determined to be the highest one of the threshold voltages $V_{\text{sub.1}}$ and $V_{\text{sub.2}}$ that was crossed by the sense voltage $V_{\text{sub.sense}}$, the reference current $I_{\text{sub.ref}}(2)$ will be equal to $2 * (N_{\text{sub.max}} * I_{\text{sub.unit}} / C_{\text{sub.ratio}})$ in accordance with the equation (Eq. 2).

[0041] In the equation (Eq. 2), $N_{\text{sub.max}}$ represents a maximum value that can be stored in an L -bit binary word, which is used to output the digital average value $V_{\text{sub.AVG}}$ and the digital peak value $V_{\text{sub.PEAK}}$, respectively. For example, if the binary word is an 8-bit binary word ($L=8$), then the $N_{\text{sub.max}}$ would be equal to 256 ($2^{\text{sup.8}}$). In this regard, $N_{\text{sub.max}}$ also corresponds to a full-scale (maximum level) of the electrical current $I_{\text{sub.actual}}$ (hereinafter referred to as “ $\text{MAX}(I_{\text{sub.actual}})$ ”) that can be handled by the SIDO ADC 12. Accordingly, it is possible to divide the full-scale electrical current $I_{\text{sub.actual}}$ into a number of bitwise units $I_{\text{sub.unit}}$ based on equation (Eq. 3) below.

$$[00003] I_{\text{unit}} = \text{MAX}(I_{\text{actual}}) / N_{\text{max}} = \text{MAX}(I_{\text{actual}}) / 2^L \quad (\text{Eq. 3})$$

[0042] The digital average voltage circuit 16 will repeat the procedures described above until an end of a predefined measurement period $T_{\text{sub.measure}}$ (not shown). In a non-limiting example, the predefined measurement period $T_{\text{sub.measure}}$ can be determined by equation (Eq. 4) below.

$$[00004] T_{\text{measure}} = N_{\text{max}} * T_{\text{clk}} = 2^L * T_{\text{clk}} \quad (\text{Eq. 4})$$

[0043] At the end of the predefined measurement period $T_{\text{sub.measure}}$, the average calculation circuit 34 will calculate a weighted average of the respective value stored in each of the first counters 32(1)-32(N) to thereby determine the digital average value $V_{\text{sub.AVG}}$. In an embodiment, the average calculation circuit 34 can determine the digital average value $V_{\text{sub.AVG}}$ based on equation (Eq. 5) below.

$$[00005] V_{\text{AVG}} = \text{Math.}_{i=1}^N i * \text{CNT}(i) \quad (\text{Eq. 5})$$

[0044] In the equation (Eq. 5), i represents an index number of the first counters 32(1)-32(N) ($1 \leq i \leq N$) and $\text{CNT}(i)$ represents the respective value stored in a respective one of the first counters 32(1)-32(N). In this regard, the weighting factor for determining the digital average value $V_{\text{sub.AVG}}$ is equal to the index number of the first counters 32(1)-32(N).

[0045] FIG. 5 is a graphic diagram illustrating a working example of the SIDO ADC 12 of FIG. 3 in determining the digital average value $V_{\text{sub.AVG}}$. Common elements between FIGS. 3, 4, and 5 are shown therein with common element numbers and will not be re-described herein. For the sake of illustration, FIG. 5 illustrates a scenario wherein the SIDO ADC 12 of FIG. 3 is configured to operate with the baseline voltage $V_{\text{sub.0}}$ and two threshold voltages $V_{\text{sub.1}}$, $V_{\text{sub.2}}$. It should be appreciated that the operating principles described herein can be applicable to any suitable number of the threshold voltages $V_{\text{sub.1}}-V_{\text{sub.N}}$ as discussed in reference to FIG. 3.

[0046] At time t_0 , which corresponds to a start of the predefined measurement period $T_{\text{sub.measure}}$, the sense voltage $V_{\text{sub.sense}}$ is equal to the baseline voltage $V_{\text{sub.0}}$. At time $t_{\text{sub.1}}$, the capacitor 22 is charged by the sense current $I_{\text{sub.sense}}$ to raise the sense voltage $V_{\text{sub.sense}}$ to the first voltage threshold $V_{\text{sub.1}}$. In a non-limiting example, the first voltage threshold $V_{\text{sub.1}}$ can be determined based on equation (Eq. 6) below.

$$[00006] V_1 = V_0 + (N_{\text{max}} * I_{\text{unit}} * T_{\text{clk}}) / (C_0 * C_{\text{ratio}}) \quad (\text{Eq. 6})$$

[0047] Accordingly, each of the threshold voltages $V_{\text{sub.1}}-V_{\text{sub.N}}$ may be defined as a multiple of the first threshold voltage $V_{\text{sub.1}}$, as in equation (Eq. 7) below.

$$[00007] V_i = i * [V_0 + (N_{\text{max}} * I_{\text{unit}} * T_{\text{clk}}) / (C_0 * C_{\text{ratio}})] \quad (\text{Eq. 7})$$

[0048] In the equation (Eq. 7), i ($1 \leq i \leq N$) represents the index number of the threshold voltages $V_{sub.1}$ - $V_{sub.N}$. As the sense voltage $V_{sub.sense}$ reaches the threshold voltage $V_{sub.1}$, the first counter **32(1)** is triggered by the first latch **30(1)** to increase by 1. Concurrently or subsequently, the switch SW is closed to couple the reference current generator **24** to the capacitor **22** to generate the reference current $I_{sub.ref}$. As discussed earlier, the reference current $I_{sub.ref}$ flows in the opposite direction relative to the sense current $I_{sub.sense}$ to offset the sense current $I_{sub.sense}$. As a result, at time $t_{sub.2}$, the sense voltage $V_{sub.sense}$ may be reduced from the threshold voltage $V_{sub.1}$ by the offset value $OFF_{sub.1}$ to a reduced voltage level $V'_{sub.1}$ ($V_{sub.0} < V'_{sub.1} < V_{sub.1}$). In a non-limiting example, the duration between time $t_{sub.1}$ and $t_{sub.2}$ can equal the respective clock cycle duration T_{clk} of each of the clock cycles **36**.

[0049] At time $t_{sub.2}$, the switch SW is opened to remove the reference current $I_{sub.ref}$. As a result, the sense voltage $V_{sub.sense}$ may start rising once again. For example, at time $t_{sub.3}$, the sense voltage $V_{sub.sense}$ may have climbed to the higher threshold voltage $V_{sub.2}$. Accordingly, the first counter **32(2)** will be triggered by the first latch **30(2)** to increase by 1. Once again, the switch SW is closed to couple the reference current generator **24** to the capacitor **22** to generate the reference current $I_{sub.ref}$ to reduce the sense current $I_{sub.sense}$ and, accordingly, the sense voltage $V_{sub.sense}$. As a result, at time t_4 , the sense voltage $V_{sub.sense}$ may be reduced from the threshold voltage $V_{sub.2}$ by the respective offset value $OFF_{sub.2}$ to a reduced voltage level $V'_{sub.2}$ ($V_{sub.1} < V'_{sub.2} < V_{sub.2}$).

[0050] In this regard, the sense voltage $V_{sub.sense}$ will increase and decrease during the predefined measurement period $T_{sub.measure}$ in a zig-zag fashion and each of the first counters **32(1)**-**32(N)** is configured to count each occurrence of the sense voltage $V_{sub.sense}$ reaching a respective one of the threshold voltages $V_{sub.1}$ - $V_{sub.N}$. At the end of the predefined measurement period (e.g., at time $t_{sub.x}$), the average calculation circuit **34** can calculate the digital average value $V_{sub.AVG}$ in accordance with the equation (Eq. 5).

[0051] With reference back to FIG. 3, the digital peak voltage circuit **18** is configured to output the digital peak value $V_{sub.PEAK}$, concurrent to the digital average voltage circuit **16** outputting the digital average value $V_{sub.AVG}$. In this regard, the digital peak voltage circuit **18** can be configured to include a clock counter **44**, multiple second counters **46(1)**-**46(N)**, and a peak calculation circuit **48**. Like the range detection logic **26** and the first latches **30(1)**-**30(N)**, the clock counter **44** is also configured to operate based on the clock signal Clk.

[0052] Herein, each of the second counters **46(1)**-**46(N)** is configured to operate synchronously with a respective one of the first latches **30(1)**-**30(N)**. Specifically, each of the first latches **30(1)**-**30(N)** is configured to generate a respective one of multiple latch signals $LM_{sub.1}$ - $LM_{sub.N}$ in response to receiving the latch signal **40** to latch a respective one of the second counters **46(1)**-**46(N)**. Together, the second counters **46(1)**-**46(N)** will record a respective duration of the sense voltage $V_{sub.sense}$ staying in each of the threshold regions $R_{sub.1}$ - $R_{sub.N}$ during the predefined measurement period $T_{sub.measure}$. At the end of the predefined measurement period $T_{sub.measure}$, the peak calculation circuit **48** will calculate the digital peak value $V_{sub.PEAK}$ based on a respective value in each of the second counters **46(1)**-**46(N)**. In an embodiment, the peak calculation circuit **48** may calculate the digital peak value $V_{sub.PEAK}$ based on equation (Eq. 8) below.

$$[00008] V_{PEAK} = \text{MAX}[i * \text{CNT}(i) / \text{MNT}(i)] (1 \leq i \leq N) \quad (\text{Eq. 8})$$

[0053] In the equation (Eq. 8), $\text{CNT}(i)$ represents a respective value in each of the first counters **32(1)**-**32(N)** and $\text{MNT}(i)$ represents a respective value in each of the second counters **46(1)**-**46(N)**.

[0054] The SIDO ADC **12** of FIGS. 2 and 3 can be provided in a user element (e.g., a wireless device) to support the embodiments described above. In this regard, FIG. 6 is a schematic diagram of an exemplary user element **100** wherein the SIDO ADC **12** of FIGS. 2 and 3 can be provided.

[0055] Herein, the user element **100** can be any type of user elements, such as mobile terminals,

smart watches, tablets, computers, navigation devices, access points, and like wireless communication devices that support wireless communications, such as cellular, wireless local area network (WLAN), Bluetooth, and near field communications. The user element **100** will generally include a control system **102**, a baseband processor **104**, transmit circuitry **106**, receive circuitry **108**, antenna switching circuitry **110**, multiple antennas **112**, and user interface circuitry **114**. In a non-limiting example, the control system **102** can be a field-programmable gate array (FPGA), as an example. In this regard, the control system **102** can include at least a microprocessor(s), an embedded memory circuit(s), and a communication bus interface(s). The receive circuitry **108** receives radio frequency signals via the antennas **112** and through the antenna switching circuitry **110** from one or more base stations. A low noise amplifier and a filter cooperate to amplify and remove broadband interference from the received signal for processing. Downconversion and digitization circuitry (not shown) will then downconvert the filtered, received signal to an intermediate or baseband frequency signal, which is then digitized into one or more digital streams using an analog-to-digital converter(s) (ADC).

[0056] The baseband processor **104** processes the digitized received signal to extract the information or data bits conveyed in the received signal. This processing typically comprises demodulation, decoding, and error correction operations, as will be discussed in greater detail below. The baseband processor **104** is generally implemented in one or more digital signal processors (DSPs) and application specific integrated circuits (ASICs).

[0057] For transmission, the baseband processor **104** receives digitized data, which may represent voice, data, or control information, from the control system **102**, which it encodes for transmission. The encoded data is output to the transmit circuitry **106**, where a digital-to-analog converter(s) (DAC) converts the digitally encoded data into an analog signal and a modulator modulates the analog signal onto a carrier signal that is at a desired transmit frequency or frequencies. A power amplifier will amplify the modulated carrier signal to a level appropriate for transmission, and deliver the modulated carrier signal to the antennas **112** through the antenna switching circuitry **110**. The multiple antennas **112** and the replicated transmit and receive circuitries **106**, **108** may provide spatial diversity. Modulation and processing details will be understood by those skilled in the art.

[0058] The SDO ADC **12** of FIG. 2 may be provided in any one or more of the circuitries in the user element **100**, including but not limited to the control system **102**, the baseband processor **104**, the transmit circuitry **106**, and the receive circuitry **108**. It should be appreciated that the SDO ADC **12** may also be provided in other types of electrical and/or electronic devices.

[0059] The SDO ADC **12** of FIG. 2 may be operated based on a process.

[0060] In this regard, FIG. 7 is a flowchart of an exemplary process **200** for operating the SDO ADC **12** of FIG. 2.

[0061] Herein, the process **200** includes receiving the analog input voltage $V_{sub.A}$ and scaling the analog input voltage $V_{sub.A}$ down to the sense voltage $V_{sub.sense}$ (step **202**). The process **200** also includes storing the baseline voltage $V_{sub.0}$ and the threshold voltages $V_{sub.1}$ - $V_{sub.N}$ each higher than the baseline voltage $V_{sub.0}$ to thereby establish the threshold regions $R_{sub.1}$ - $R_{sub.N}$ (step **204**). The process **200** also includes counting each occurrence of the sense voltage $V_{sub.sense}$ being higher than or equal to a highest one of the threshold voltages $V_{sub.1}$ - $V_{sub.N}$ that falls below the sense voltage $V_{sub.sense}$ during the predefined measurement period $T_{sub.measure}$ in a corresponding one of the first counters **32(1)**-**32(N)** (step **206**). The process **200** also includes reducing the sense voltage $V_{sub.sense}$ by a respective one of the offset values $OFF_{sub.1}$ - $OFF_{sub.N}$ corresponding to the highest one of the threshold voltages $V_{sub.1}$ - $V_{sub.N}$ that falls below the sense voltage $V_{sub.sense}$ (step **208**). The process **200** also includes outputting the digital average value $V_{sub.AVG}$ indicating the average of the analog input voltage $V_{sub.A}$ during the predefined measurement period $T_{sub.measure}$ based on a respective value in each of the first counters **32(1)**-**32(N)** (step **210**). The process **200** also includes counting a respective duration

of the sense voltage $V_{sub.sense}$ staying within each of the threshold regions $R_{sub.1}$ - $R_{sub.N}$ during the predefined measurement period $T_{sub.measure}$ in multiple second counters **46(1)-46(N)** (step **212**). The process **200** also includes outputting the digital peak value $V_{sub.PEAK}$ indicating a peak of the analog input voltage $V_{sub.A}$ during the predefined measurement period $T_{sub.measure}$ based on a respective value in each of the first counters **32(1)-32(N)** and each of the second counters **46(1)-46(N)** (step **214**).

[0062] Those skilled in the art will recognize improvements and modifications to the preferred embodiments of the present disclosure. All such improvements and modifications are considered within the scope of the concepts disclosed herein and the claims that follow.

Claims

1. A single-input dual-output (SIDO) analog-to-digital converter (ADC) comprising: a scaling circuit configured to receive an analog input voltage and scale the analog input voltage down to a sense voltage; a digital average voltage circuit configured to: store a baseline voltage and a plurality of threshold voltages each higher than the baseline voltage to thereby establish a plurality of threshold regions; count each occurrence of the sense voltage being higher than or equal to a highest one of the plurality of threshold voltages that falls below the sense voltage during a predefined measurement period in a corresponding one of a plurality of first counters; reduce the sense voltage by a respective one of a plurality of offset values corresponding to the highest one of the plurality of threshold voltages that falls below the sense voltage; and output a digital average value indicating an average of the analog input voltage during the predefined measurement period based on a respective value in each of the plurality of first counters; and a digital peak voltage circuit configured to: count a respective duration of the sense voltage staying within each of the plurality of threshold regions during the predefined measurement period in a plurality of second counters; and output a digital peak value indicating a peak of the analog input voltage during the predefined measurement period based on a respective value in each of the plurality of first counters and each of the plurality of second counters.
2. The SIDO ADC of claim 1, wherein the scaling circuit comprises: a voltage-to-current converter configured to convert the analog input voltage to a sense current based on a scaling factor that is less than one; a capacitor configured to convert the sense current into the sense voltage; and a reference current generator configured to generate a reference current to offset the sense current to thereby reduce the sense voltage by the respective one of the plurality of offset values.
3. The SIDO ADC of claim 2, wherein the reference current is proportionally increased in accordance with the plurality of threshold voltages.
4. The SIDO ADC of claim 1, wherein the digital average voltage circuit comprises: a plurality of voltage comparators each configured to: compare the sense voltage with a respective one of the plurality of threshold voltages to determine whether the sense voltage is higher than or equal to the respective one of the plurality of threshold voltages; and output a respective threshold crossing indication in response to determining that the sense voltage is higher than or equal to the respective one of the plurality of threshold voltages; a range detection logic configured to: determine the respective threshold crossing indication provided by a respective one of the plurality of voltage comparators having the highest one of the plurality of threshold voltages that falls below the sense voltage; cause a respective one of the plurality of first counters corresponding to the respective one of the plurality of voltage comparators to increase by one; and cause the sense voltage to be reduced by the respective one of the plurality of offset values; and an average calculation circuit configured to determine a weighted average of the respective value in each of the plurality of first counters to thereby output the digital average value.
5. The SIDO ADC of claim 4, wherein the average calculation circuit is further configured to determine the weighted average based on a plurality of weight factors that is increased in

accordance with the plurality of threshold voltages.

6. The SIDO ADC of claim 1, wherein the digital peak voltage circuit comprises: the plurality of second counters each latched with a respective one of the plurality of first counters to count the respective duration of the sense voltage staying within each of the plurality of threshold regions; and a peak calculation circuit configured to determine the digital peak value based on a respective value in each of the plurality of second counters.

7. The SIDO ADC of claim 1, wherein each of the plurality of threshold regions corresponds to an identical voltage differential between each pair of adjacent threshold voltages among the plurality of threshold voltages.

8. A method for converting an analog input voltage into a digital average value and a digital peak value comprising: receiving the analog input voltage and scaling the analog input voltage down to a sense voltage; storing a baseline voltage and a plurality of threshold voltages each higher than the baseline voltage to thereby establish a plurality of threshold regions; counting each occurrence of the sense voltage being higher than or equal to a highest one of the plurality of threshold voltages that falls below the sense voltage during a predefined measurement period in a corresponding one of a plurality of first counters; reducing the sense voltage by a respective one of a plurality of offset values corresponding to the highest one of the plurality of threshold voltages that falls below the sense voltage; outputting the digital average value indicating an average of the analog input voltage during the predefined measurement period based on a respective value in each of the plurality of first counters; counting a respective duration of the sense voltage staying within each of the plurality of threshold regions during the predefined measurement period in a plurality of second counters; and outputting the digital peak value indicating a peak of the analog input voltage during the predefined measurement period based on a respective value in each of the plurality of first counters and each of the plurality of second counters.

9. The method of claim 8, further comprising: converting, using a voltage-to-current converter, the analog input voltage to a sense current based on a scaling factor that is less than one; converting, using a capacitor, the sense current into the sense voltage; and generating, using a reference current generator, a reference current to offset the sense current to thereby reduce the sense voltage by the respective one of the plurality of offset values.

10. The method of claim 9, further comprising increasing the reference current proportionally in accordance with the plurality of threshold voltages.

11. The method of claim 8, further comprising: comparing, using each of a plurality of voltage comparators, the sense voltage with a respective one of the plurality of threshold voltages to determine whether the sense voltage is higher than or equal to the respective one of the plurality of threshold voltages; outputting, from each of the plurality of voltage comparators, a respective threshold crossing indication in response to determining that the sense voltage is higher than or equal to the respective one of the plurality of threshold voltages; determining, using a range detection logic, the respective threshold crossing indication provided by a respective one of the plurality of voltage comparators having the highest one of the plurality of threshold voltages that falls below the sense voltage; causing, by the range detection logic, a respective one of the plurality of first counters corresponding to the respective one of the plurality of voltage comparators to increase by one; causing, by the range detection logic, the sense voltage to be reduced by the respective one of the plurality of offset values; and determining, using an average calculation circuit, a weighted average of the respective value in each of the plurality of first counters to thereby output the digital average value.

12. The method of claim 11, further comprising determining, using the average calculation circuit, the weighted average based on a plurality of weight factors that is increased in accordance with the plurality of threshold voltages.

13. The method of claim 8, further comprising: latching each of the plurality of second counters with a respective one of the plurality of first counters to count the respective duration of the sense

voltage staying within each of the plurality of threshold regions; and determining, using a peak calculation circuit, the digital peak value based on a respective value in each of the plurality of second counters.

14. The method of claim 8, further comprising defining each of the plurality of threshold regions to correspond to an identical voltage differential between each pair of adjacent threshold voltages among the plurality of threshold voltages.

15. A wireless device comprising at least one single-input dual-output (SIDO) analog-to-digital converter (ADC), comprising: a scaling circuit configured to receive an analog input voltage and scale the analog input voltage down to a sense voltage; a digital average voltage circuit configured to: store a baseline voltage and a plurality of threshold voltages each higher than the baseline voltage to thereby establish a plurality of threshold regions; count each occurrence of the sense voltage being higher than or equal to a highest one of the plurality of threshold voltages that falls below the sense voltage during a predefined measurement period in a corresponding one of a plurality of first counters; reduce the sense voltage by a respective one of a plurality of offset values corresponding to the highest one of the plurality of threshold voltages that falls below the sense voltage; and output a digital average value indicating an average of the analog input voltage during the predefined measurement period based on a respective value in each of the plurality of first counters; and a digital peak voltage circuit configured to: count a respective duration of the sense voltage staying within each of the plurality of threshold regions during the predefined measurement period in a plurality of second counters; and output a digital peak value indicating a peak of the analog input voltage during the predefined measurement period based on a respective value in each of the plurality of first counters and each of the plurality of second counters.

16. The wireless device of claim 15, wherein the at least one SIDO ADO is provided in one or more of a control system, a baseband processor, transmit circuitry, and receive circuitry in the wireless device.

17. The wireless device of claim 15, wherein the scaling circuit comprises: a voltage-to-current converter configured to convert the analog input voltage to a sense current based on a scaling factor that is less than one; a capacitor configured to convert the sense current into the sense voltage; and a reference current generator configured to generate a reference current to offset the sense current to thereby reduce the sense voltage by the respective one of the plurality of offset values.

18. The wireless device of claim 15, wherein the digital average voltage circuit comprises: a plurality of voltage comparators each configured to: compare the sense voltage with a respective one of the plurality of threshold voltages to determine whether the sense voltage is higher than or equal to the respective one of the plurality of threshold voltages; and output a respective threshold crossing indication in response to determining that the sense voltage is higher than or equal to the respective one of the plurality of threshold voltages; a range detection logic configured to: determine the respective threshold crossing indication provided by a respective one of the plurality of voltage comparators having the highest one of the plurality of threshold voltages that falls below the sense voltage; cause a respective one of the plurality of first counters corresponding to the respective one of the plurality of voltage comparators to increase by one; and cause the sense voltage to be reduced by the respective one of the plurality of offset values; and an average calculation circuit configured to determine a weighted average of the respective value in each of the plurality of first counters to thereby output the digital average value.

19. The wireless device of claim 15, wherein the digital peak voltage circuit comprises: the plurality of second counters each latched with a respective one of the plurality of first counters to count the respective duration of the sense voltage staying within each of the plurality of threshold regions; and a peak calculation circuit configured to determine the digital peak value based on a respective value in each of the plurality of second counters.

20. The wireless device of claim 15, wherein each of the plurality of threshold regions corresponds

to an identical voltage differential between each pair of adjacent threshold voltages among the plurality of threshold voltages.
